Jason Wakim Richards

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11 August 2026

Astrolab
Software Engineering
Hawthorne, California

Application for GNC Engineer

Dear Denise Brown and Hiring Team,

I am applying for the GNC Engineer position at Astrolab because building the systems required for sustained lunar operations is exactly the direction in which I have developed my career. My background combines planetary-rover GNC, spacecraft attitude determination and control, and hands-on autonomous UAV development. I am particularly motivated by the opportunity to design, analyze, tune, and validate GNC algorithms for crewed, semi-autonomous, and fully autonomous rovers, then prove their performance in simulation, integrated testbeds, and test vehicles.

At Airbus Defence & Space, I worked in the AOCS/GNC team on autonomous planetary-rover technology and served as lead test engineer for LiDAR activities during Integrated BreadBoarding 3 and 4. I planned test procedures and supported end-to-end verification of a Sample Fetch Rover concept at Technology Readiness Level 6. I designed a networked hardware/software test bench, integrated LiDAR with rover breadboards through ROS, collected campaign data, characterized sensor performance, and validated navigation models. I also developed C/C++ software on Linux for an offline GNC simulation that generated synthetic camera imagery and LiDAR scans, and investigated terrain generation and Vector Field Histogram planning for lunar rovers.

I now develop and deploy autonomous UAV systems in the Autonomous Robotics Group in Luxembourg using C++, Python, ROS 2, PX4, embedded Linux, cameras, IMUs, OpenVINS, Docker, and Kalibr. My work spans localization, Semantic Relational SLAM, planning, control, camera–IMU calibration, simulation, laboratory integration, and flight testing. Previously, as ADCS Lead for the JamSail 3U CubeSat, I defined requirements and interfaces, developed embedded C flight software and a C/C++ simulation, and implemented an EKF that fused sun-sensor, magnetometer, and IMU measurements. I also performed statistical analysis on sensors to charecterise the noise parameters through fourier analysis. This then enabled me to tune the ADCS filters to reduce determination noise within the system and achieve a sub 5 degree determination error with just a sun sensor, magnetometer and IMU. Performace was verified through development of an N-Body Simulator which was developed in line with the European ECSS standards.

This experience maps closely to Astrolab's need for engineers who own GNC from requirements and algorithm prototypes through integrated verification and validation. I bring practical experience with Python and C++, GNC simulations, EKF-based state estimation, control algorithms, IMUs, cameras, LiDAR, sensor characterization, calibration, performance analysis, and vehicle testing. I am comfortable translating mission needs into traceable requirements and test plans, documenting algorithms and interfaces, and working across software, sensing, autonomy, and hardware teams to resolve integration issues. As a UKSEDS Competitions Management Co-Lead, I also coordinate volunteers and industrial partners across four national competitions, reinforcing my ability to take ownership and deliver reliably against fixed milestones.

I would be proud to bring this combination of lunar-rover development, embedded flight software, autonomous-vehicle testing, and spacecraft GNC to Astrolab. I share your focus on turning robust engineering into dependable capability, and I would welcome the opportunity to help mature and prove the GNC systems on which sustained lunar operations will depend.

Yours sincerely,

Jason Wakim Richards